

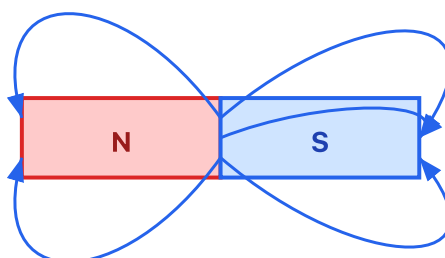
Topic 12 — Magnetism and the Motor Effect

Edexcel GCSE (9-1) Physics (1PH0) — Paper 2

12.1 Magnets and Fields

- Every magnet has a **north** and a **south** pole.
- Like poles repel; unlike poles attract.
- Field lines go from **N to S** outside the magnet.
- Ferromagnetic materials: **iron, nickel, cobalt, steel**.

Field around a bar magnet



Lines emerge at N, curve round, enter S; closer lines = stronger field.

12.2 Permanent vs Induced Magnets

- **Permanent magnet** — keeps its magnetism (e.g. hard steel).
- **Induced magnet** — becomes a magnet only when placed in a field; always experiences attraction; loses most magnetism when removed (e.g. soft iron paperclip).

12.3 Electromagnets

A coil (solenoid) with current through it produces a magnetic field similar to a bar magnet. Adding a **soft iron core** makes it much stronger. Strength increases with:

- More current;
- More turns per unit length;
- Using a soft iron core.

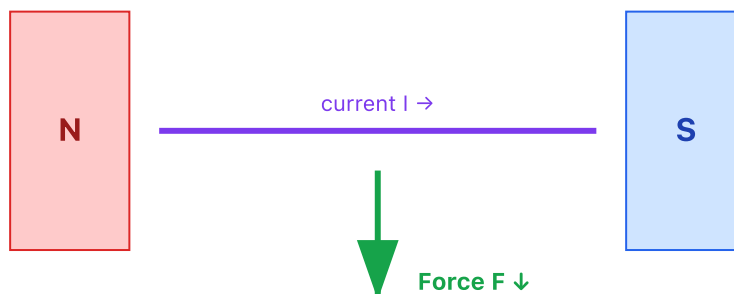
12.4 The Motor Effect

A current-carrying conductor in a magnetic field experiences a force. The force is **perpendicular** to both the current and the field, given by Fleming's **left-hand rule**.

$$F = B \times I \times L \quad (\text{N, T, A, m})$$

Fleming's Left-Hand Rule

Thumb = Force, First finger = Field, Second = Current



Worked Example: A 20 cm wire carries 3 A perpendicular to a 0.40 T field.

$F = BIL = 0.40 \times 3 \times 0.20 = \mathbf{0.24 \text{ N}}$, perpendicular to both the field and the current.

12.5 The DC Motor

A rectangular coil in a magnetic field carries a current. Each side of the coil feels a force in opposite directions, producing a turning moment. A **split-ring commutator** reverses the current

direction every half-turn so the coil keeps spinning the same way.

12.6 Loudspeakers

AC through a coil in a magnetic field produces a varying force on the coil, which vibrates a cone and produces sound of the same frequency as the signal.

Key Points Summary

- Every magnet has N and S poles; like repel, unlike attract.
- Electromagnets: solenoid + soft iron core; strength varies with I and N.
- Motor effect: $F = BIL$ on a current-carrying wire in a field.
- Fleming's LH rule gives force direction in motors.
- DC motor uses a split-ring commutator to keep spin direction constant.